Aman Kumar

AI Engineer | IIT Roorkee Graduate '26
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Professional Summary

IIT Roorkee graduate (Integrated Dual Degree (B.Tech + M.Tech), 2026) who builds production-style AI systems end to end: multi-agent workflows (LangGraph), RAG with hybrid retrieval (FAISS + BM25 + cross-encoder reranking), and LLM applications served through FastAPI and React. Evaluates what gets built: retrieval benchmarks (hit@k, MRR), token and latency accounting, verifier nodes that reject hallucinated output, and server-side citation validation. Interned at an AI startup owning a production RAG and classification pipeline end to end. Competitive programmer with Codeforces 1550 (Specialist) and CodeChef 2100 (5-Star).

Education

B.Tech + M.Tech (Integrated Dual Degree), Geophysical Technology | Indian Institute of Technology, Roorkee | 2026

Intermediate (Class XII) | Gaya College (BSEB), Gaya, Bihar | 2021

Matriculate (Class X) | B.L Indo Anglian Public School (CBSE) | 2019

Technical Skills

Agentic Frameworks: LangGraph, LangChain, Multi-Agent Orchestration, Tool Calling, Structured Outputs, Verifier and Guardrail Nodes

RAG & Retrieval: FAISS, BM25, Hybrid Search (RRF), Cross-Encoder Reranking, Knowledge Graphs (NetworkX), Embeddings, Citation Grounding

LLM Stack: Groq, OpenAI API, Claude API, HuggingFace Transformers, Prompt Engineering, Token Streaming (SSE)

Evaluation & Ops: Retrieval Benchmarks (hit@k, MRR), Token/Latency Accounting, Pytest, Fake-LLM Testing, Docker

Languages & Backend: Python, TypeScript, JavaScript, C++, FastAPI, Flask, REST APIs, WebSockets, Git, GitHub

Frontend & Databases: React, Tailwind CSS, SQLAlchemy, SQLite, MySQL, Vector DBs (FAISS)

Work Experience

Software Development Intern | CallSure AI **Jun 2025 – Jul 2025**

- Owned end-to-end design and deployment of a production RAG pipeline, LangChain + FAISS indexing 500+ historical resolutions, OpenAI-compatible LLM API for generation, grounded responses with context retrieval, replacing a fully manual lookup workflow with measurable reduction in resolution time.
- Built a semantic classification system using HuggingFace embeddings with structured output routing across 3+ urgency tiers; iterated on prompts, validated accuracy on held-out test cases, and wired the output into a Flask API triggering priority escalation paths in under 2 seconds.
- Delivered the full AI system solo within a 2-month window, architecture, implementation, testing, and documentation, with the output running live in production by end of internship.

Projects

Legal Document Intelligence: RAG Platform | FastAPI, React, TypeScript, FAISS, BM25, LangChain, Groq **Jan 2025 – Oct 2026**

- Built a contract question-answering platform (FastAPI, React, SSE streaming) with clause-aware chunking and hybrid retrieval: FAISS dense search and BM25 fused by reciprocal-rank fusion, then reranked by a cross-encoder; answers stream with server-validated clause citations in about 1 to 2 seconds.
- Wrote an evaluation harness comparing five retrieval strategies on 24 questions: hybrid reached 100% hit@5, reranking lifted top-1 accuracy to 0.75 (MRR 0.85), and sending only the top-5 clauses used about 77% fewer prompt tokens than full-corpus context.
- Added an LLM-extracted knowledge graph (grounded against source text to reject hallucinated entities), clause-risk screening and a React graph explorer, all fed by a background ingestion pipeline with live progress.

Autonomous Code Review Agent | FastAPI, LangGraph, React, TypeScript, FAISS, SSE, GitHub API **Feb 2025 – Oct 2026**

- Built a multi-agent code reviewer on LangGraph and FastAPI: static analysis and FAISS-retrieved guidance feed four parallel specialist agents (security, bugs, performance, maintainability), then a verifier node drops findings on nonexistent lines and merges duplicates before scoring.
- Integrated GitHub with diff parsing scoped to changed lines, HMAC-verified webhooks and inline PR comments; reviews finish in 5 to 40 seconds and degrade gracefully when the LLM is rate-limited.
- Shipped a React and TypeScript UI with live pipeline progress over SSE, a Monaco code viewer, findings triage and a quality-trend dashboard; core pipeline covered by automated tests using a fake LLM.

Achievements

- Attained Codeforces rating of 1550 (Specialist) and CodeChef rating of 2100 (5-Star); solved 1500+ algorithmic problems, rigorous analytical foundation underlying every AI system design decision.
- Delivered end-to-end working AI prototypes solo within 24-hour hackathon constraints at IIT Roorkee.
- Secured top 1% rank in JEE Mains and JEE Advanced among 1.2M+ candidates; admitted to IIT Roorkee.

Hobbies & Interests

Football & Cricket: Enjoy playing and following both sports. **Gaming:** Regular PUBG player, fast decisions and team coordination under pressure. **Tech Trends:** Actively follow frontier LLM research, agentic AI architectures, and open-source model releases through arXiv, researcher blogs, and founder/AI podcasts. **Travelling:** Like exploring new places and experiencing different cultures.